# SARS-CoV-2
# Immuno-Vaccinology

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# Systematic Review: CoV-2

**Contents:**

- Overview CoV-2
- Literature: **in-silico** 2020 – 2025
- Immuno-Informatics:
  - Epitopes, Vaccines, Immuno-Simulations
- Pharmacology
  - Docking: Drugs, Phytocompounds

# Overview: Human CoVs

## Upper Respiratory Tract (URT):

- **Alpha-CoVs:** 229E, NL63;
- **Beta-CoVs:** OC43, KHU1;
- 15% of **common colds**: autumn/winter season;
- Usually **NO** pneumonia;

## Lower Respiratory Tract (LRT):

- **Beta-CoVs:** SARS-CoV, SARS-CoV-2, MERS;

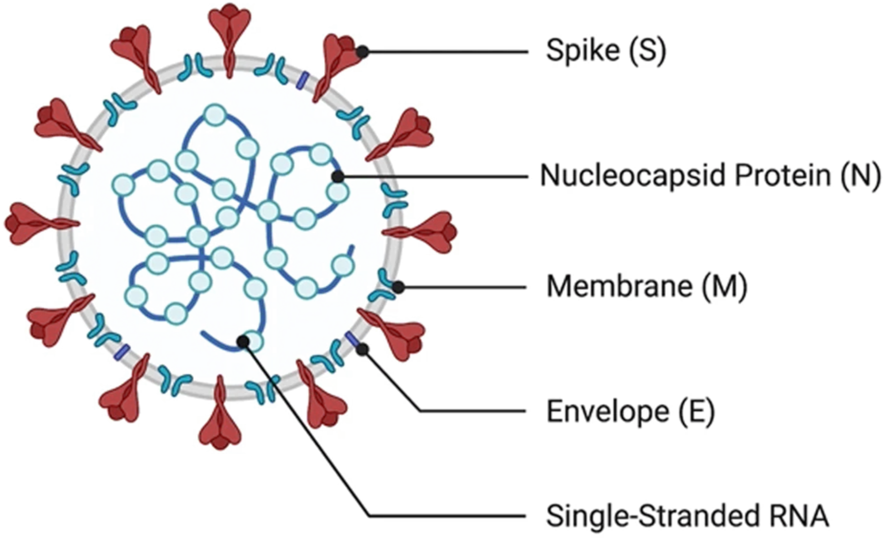
# Overview: Human CoVs

## Structure:

- EM: crown-like shape (corona);
- Enveloped;

## Surface proteins:

- **Membrane protein** (M);
- **Spike**: > 1000 AA;
- **Envelope** (E): < 100 AA;



# Overview: Human CoVs

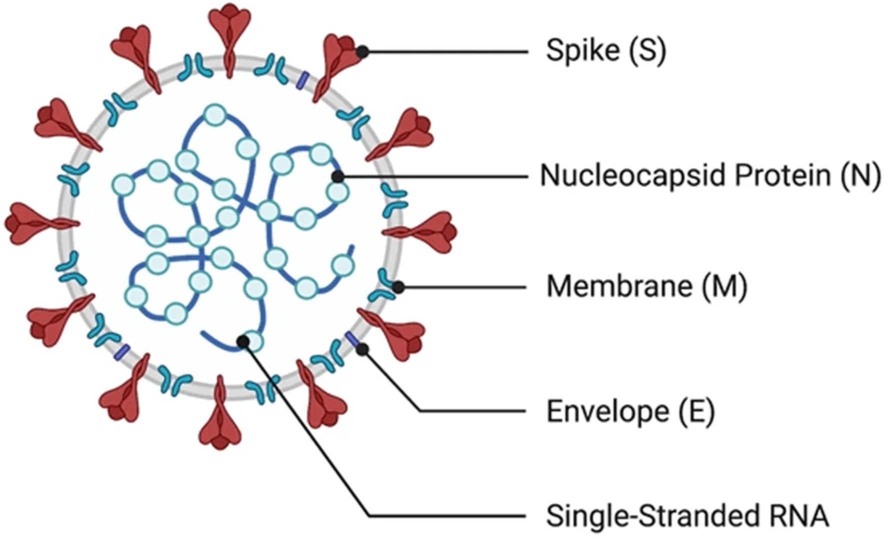
## Neutralising Antibodies:

- Only against: S, M, E; (*)
- **Cannot bind:** N, NSPs;

## Surface proteins:

- **Membrane protein** (M);
- **Spike**: > 1000 AA;
- **Envelope** (E): < 100 AA;

**Enveloped**




* E may be too small (< 100 AA); also part of it is trans-membrane;

# Overview: Human CoVs

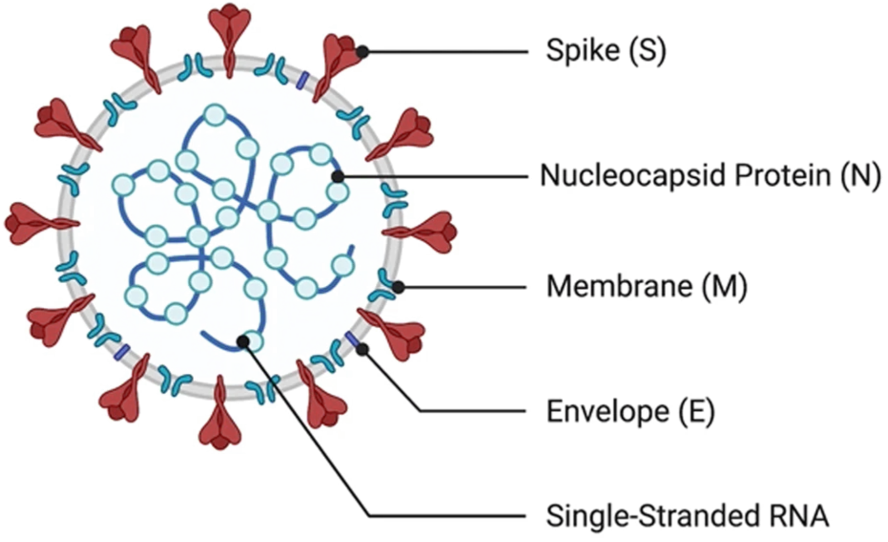
**Neutralising Antibodies:**

- Only against: **S**, **M**, **E**;
- **Cannot bind:** N, NSPs;

**Anti-N Abs:**

- Cell lysis => free N protein =>
  => Ig against **linear** N epitopes;
- NO neutralisation!
- May still be useful: Diagnostic assays; (#)

**Enveloped**




# But NOT as vaccine that induces antibodies;

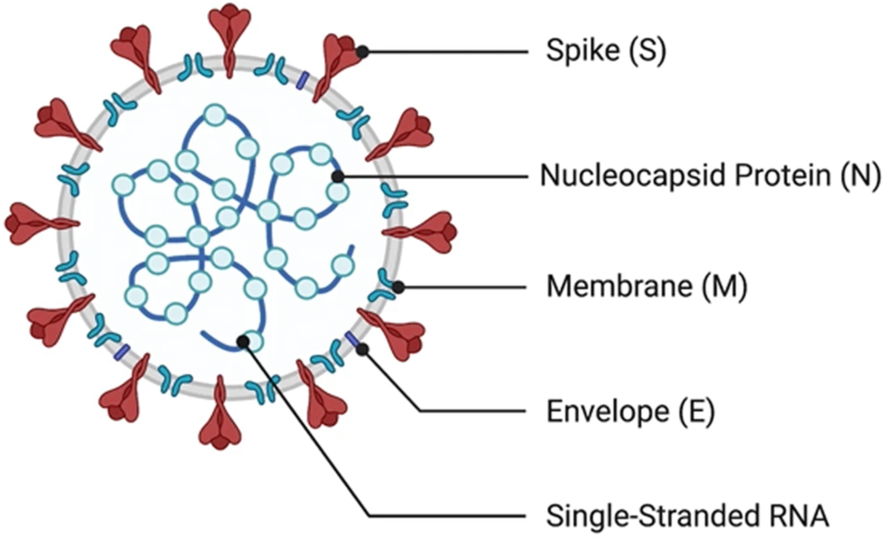
# Overview: Human CoVs

## Genome:

- Single stranded **RNA**;
- Plus-sense **RNA**;
- 30 kb (single molecule);

## Genes:

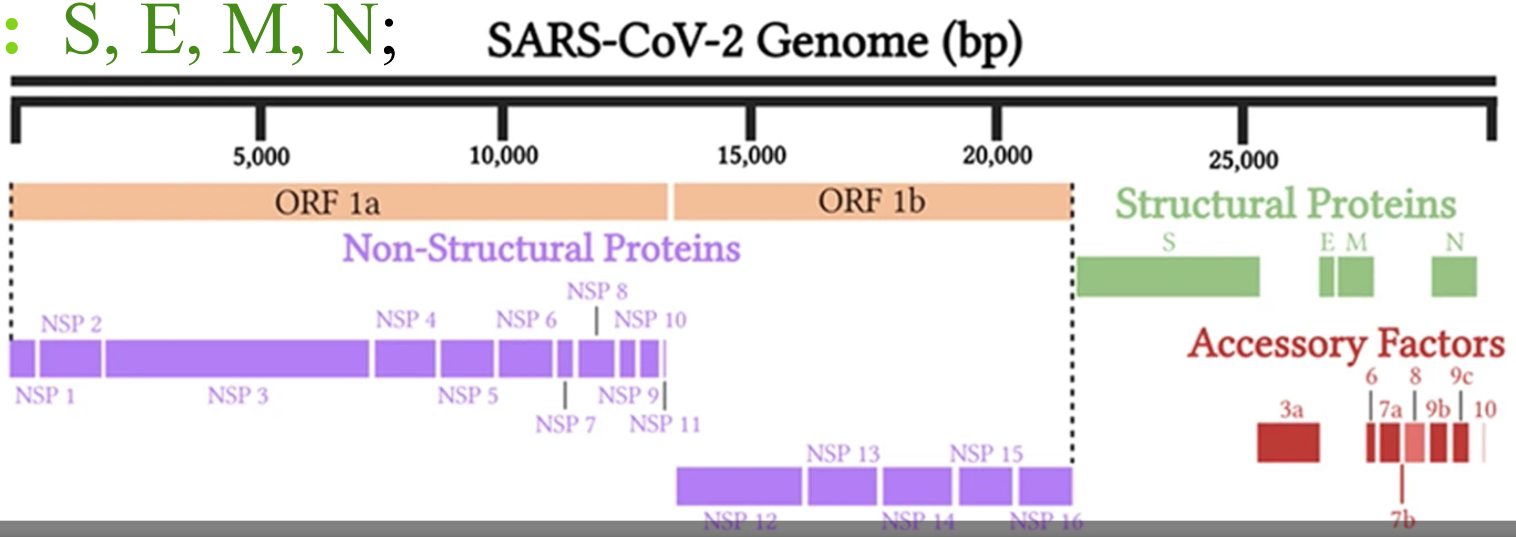
- NSPs & Structural proteins;



# Overview: SARS CoV-2

**Genome:**

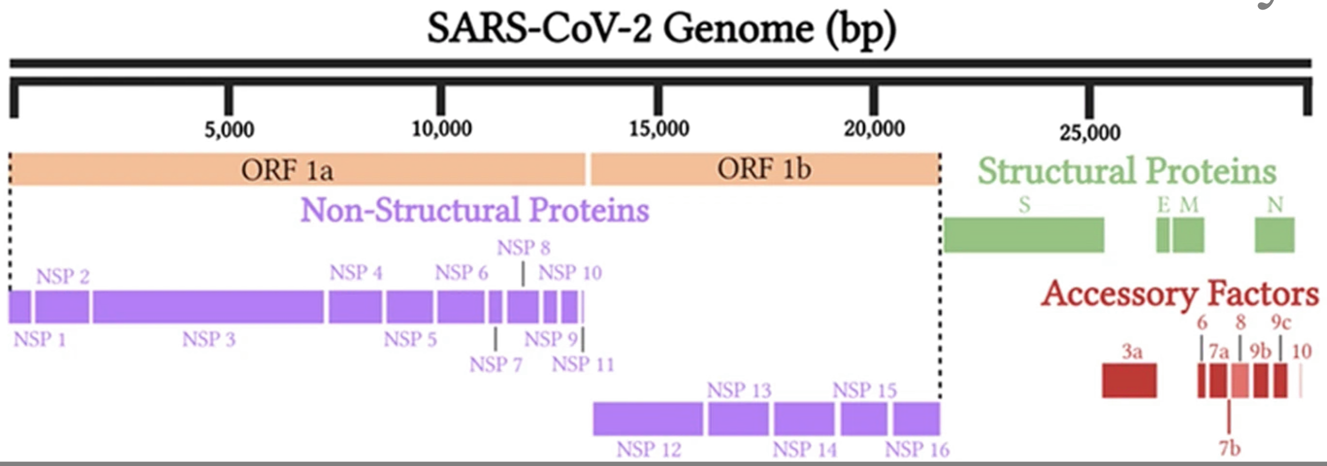
- **NSP:** ORF1a/b **Polyproteins**, others (Accessory);
- **SP:** S, E, M, N;



# Overview: SARS CoV-2

## Polyproteins: ORF1a/b

- Cleaved by **proteases**: into 10 & 5 proteins;




## Proteases:

- Main Protease: Mpro;
- Chymotrypsin-Like Pro. (CLpro)

# Literature Review: CoV-2

## PubMed Query:

- SARS CoV-2;
- Immuno-informatics or in-silico;

## Refinement:

- Manually => Pharmacology, Vaccinology, Other;
- NO loss of any articles;

# Literature Review: CoV-2 / in-silico

## PubMed Articles:

| | 2020 | 2021 | 2022 | 2023 2025 |
|---|---|---|---|---|
| **Total** | 600 | 1,128 | 1,027 | 1,242 |
| **Vaccine** | 90 | 107 | 89 | 312 |
| **Pharma** | 339 * | 450 * | 276 * | ND |
| **ND*** | 70 * | 392 * | 612 * | ND |

Difference: Other categories;

* Mostly Pharma (NO vaccine); ** ND: Not yet done;

# Literature Review: CoV-2 / in-silico

## PubMed Articles:

| | 2020 | 2021 |
|---|---|---|
| Total Vaccinology | 90 | 107 |
| Epitopes | 31 | 34 |
| Vaccines | 50 | 64 |
| Other | 9 * | 9 * |

* Reviews, Immunology (related to vaccines);

# Literature Review: CoV-2 / in-silico (only 2020)

**Target Proteins:**

- S: 23 (only S) + 11 (in combination);
- M: 12 (mostly in combination);
- Others: N (7), E, NSPs;

**Epitope types:**

- HLA-I, HLA-II;
- LB: linear & discontinuous;

# Literature Review: CoV-2 / in-silico (only 2020)

**CoV-2 Genomes:**

- Single strain: possibly most; (*)
- Consensus sequence from multiple strains;
- Example PMID 32832263:
  - Consensus: 475 strains;
  - Epitopes: 13 HLA-I & 10 HLA-II;
  - Source: S, M and other proteins;

* Not yet quantified;

# Literature Review: CoV-2 / in-silico

**Vaccine Construct:**

- CTL Epitopes;
- HTL Epitopes;
- B Epitopes;
- Adjuvants: beta-Defensin, PADRE, other sequences;
- Linkers: AAY, GPGPG, EAAAK, …;

**Example:**

- PMID: 32962156
- bDef – EAAAK – $(\text{CTL/AAY})_{35}$ – $(\text{HTL/GPGPG})_{21}$ – bDef
- Total: 1057 aa;

# Literature Review: CoV-2 / in-silico

**Docking:**

- TLR: 3 & 4 most frequent;
- Other: TLR 2,5,7,8,9 (a few);
- HLA molecules;
- Constructs: ~ 100 aa – 1057 aa; (some heroic dockings)
- Compare Hb: each chain 141 – 146 aa;

# Literature Review: CoV-2 / in-silico

**Docking:**

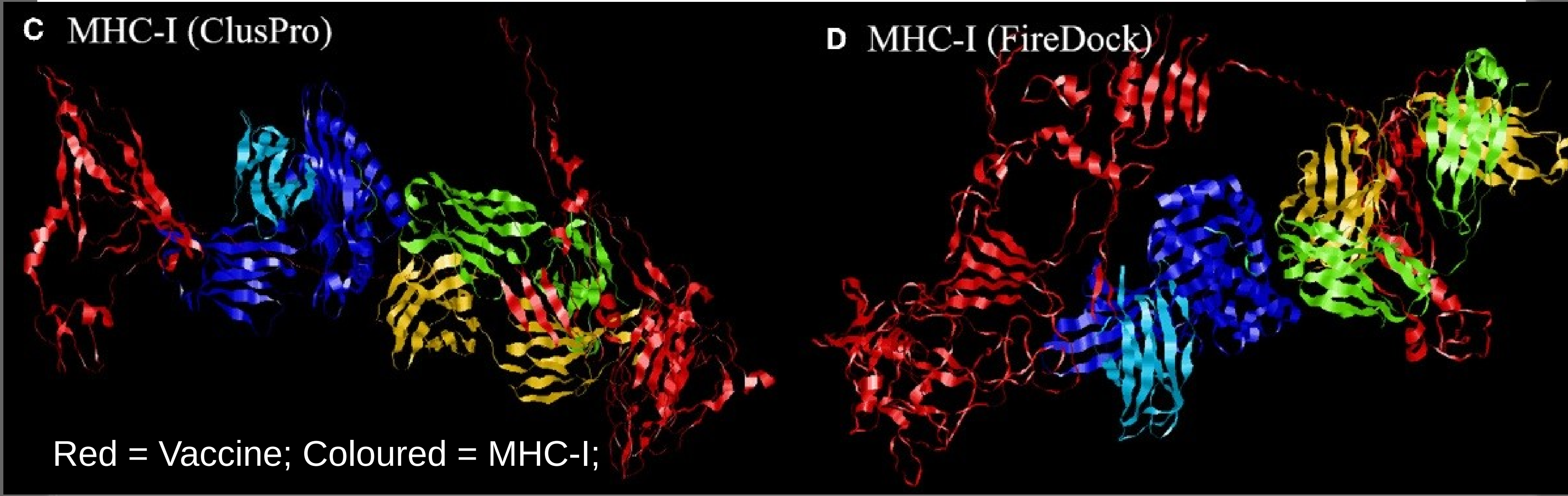
- TLRs & HLA;
- Constructs: ~ 100 aa – 1057 aa; (some **heroic dockings**)
- Compare Hb: each chain 141 – 146 aa;

**Recommendations:**

- **India & China** should probably prioritize development of infrastructures for *protein NMR;*
- build first a solid foundation for protein structures;

# Literature Review: CoV-2 / in-silico

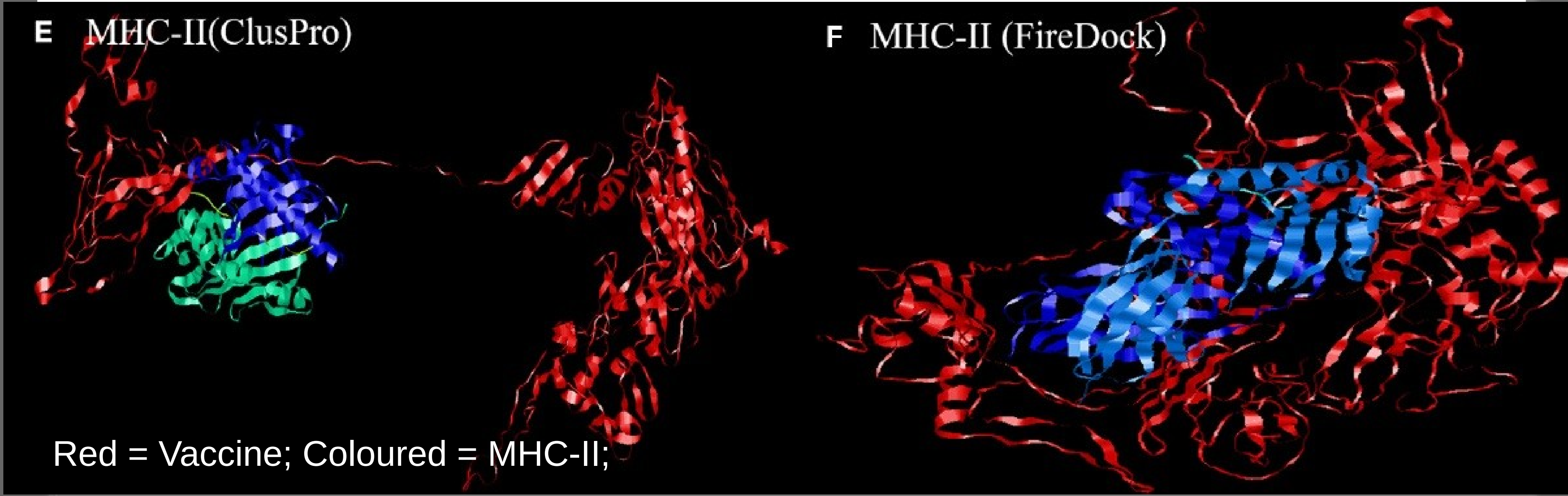
**Docking:** **MHC-I + whole protein Vaccine**




PMID: 32849643; Front. Immunol. 2020;

# Literature Review: CoV-2 / in-silico

## **Docking:** MHC-II + whole protein Vaccine

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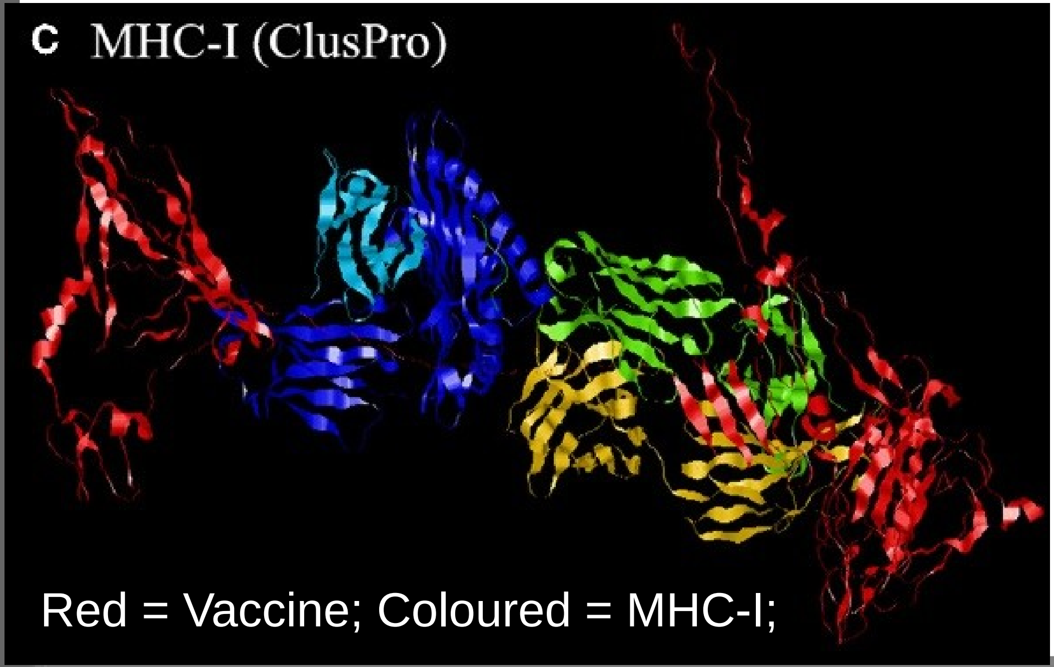
# Literature Review: CoV-2 / in-silico

## Docking: MHC-I + whole protein Vaccine




**Is it possible to dock almost any protein?**

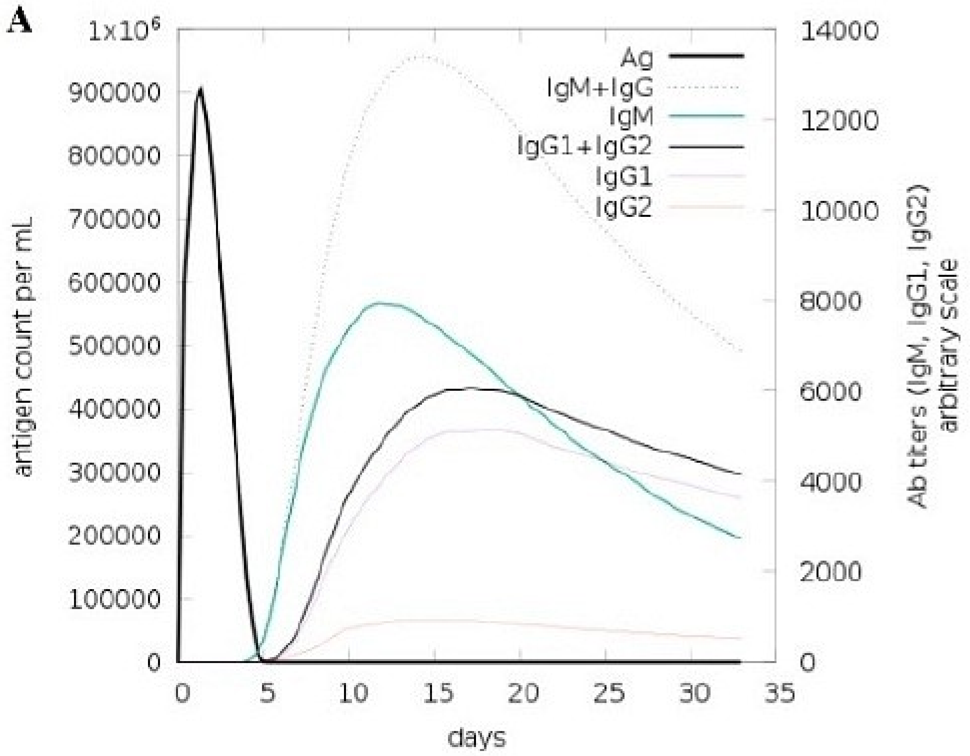
- Interesting docking exercise;
- Immunologically: absolute bollocks;
- Frontiers in Immunology!

PMID: 32849643; Front. Immunol. 2020;

# Literature Review: CoV-2 / in-silico

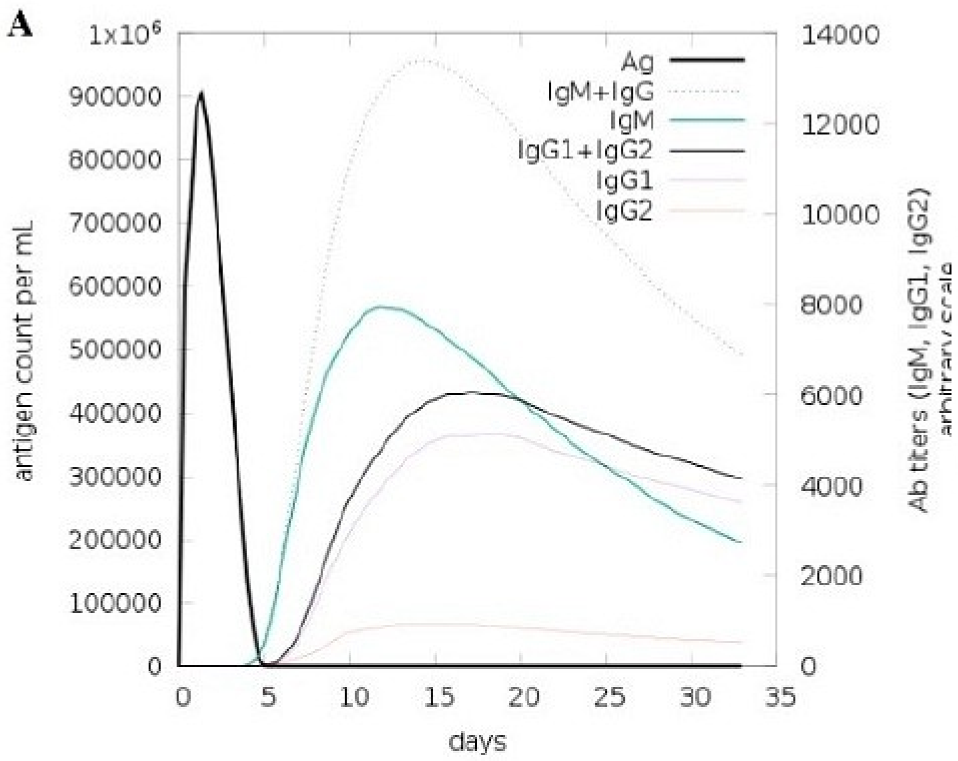
## Immune Simulations:

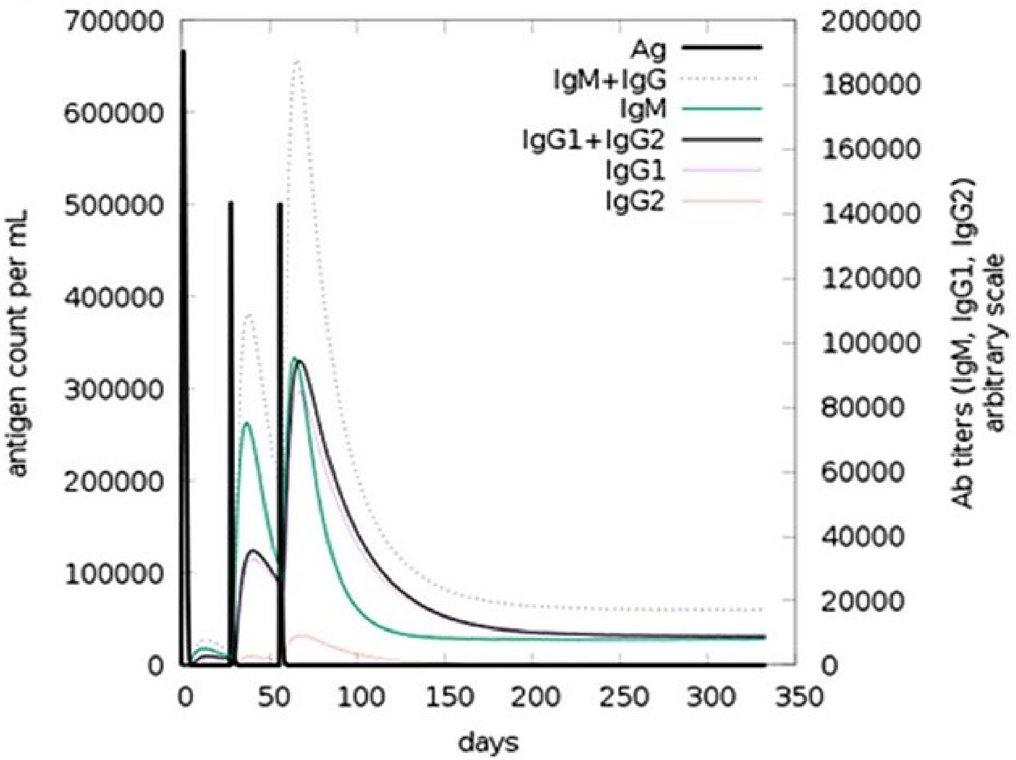
- C-ImmSim 10.1 (2020)
- Ig sub-classes
- Cells: Tc, Th, LB
- Cytokines



# Literature Review: CoV-2 / in-silico

## **Immune Simulations: Antibodies**

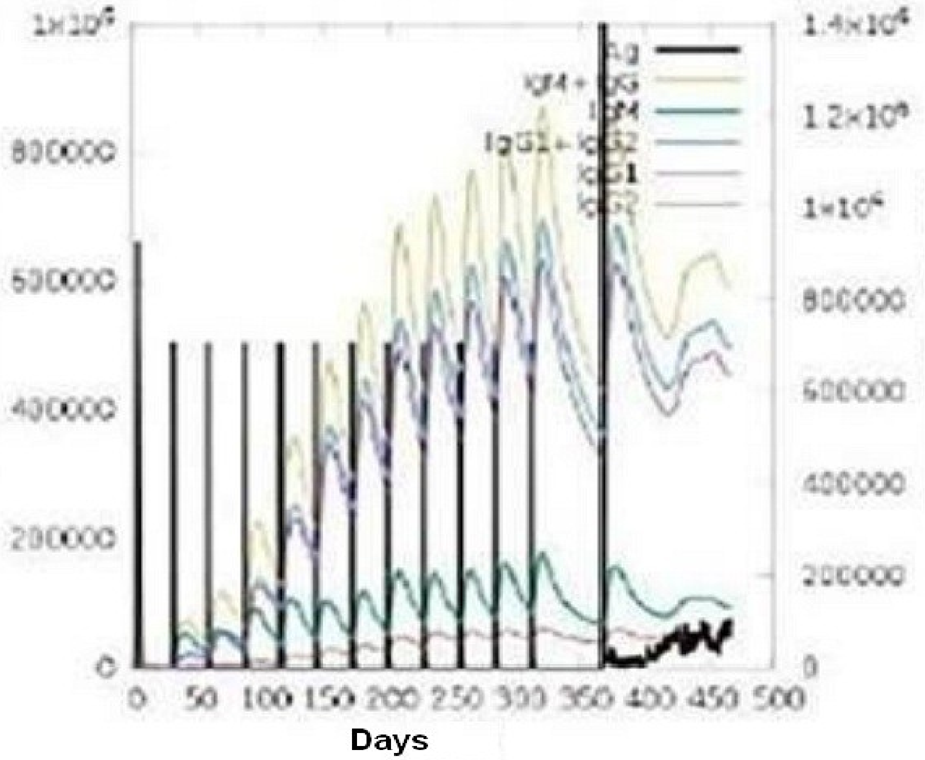


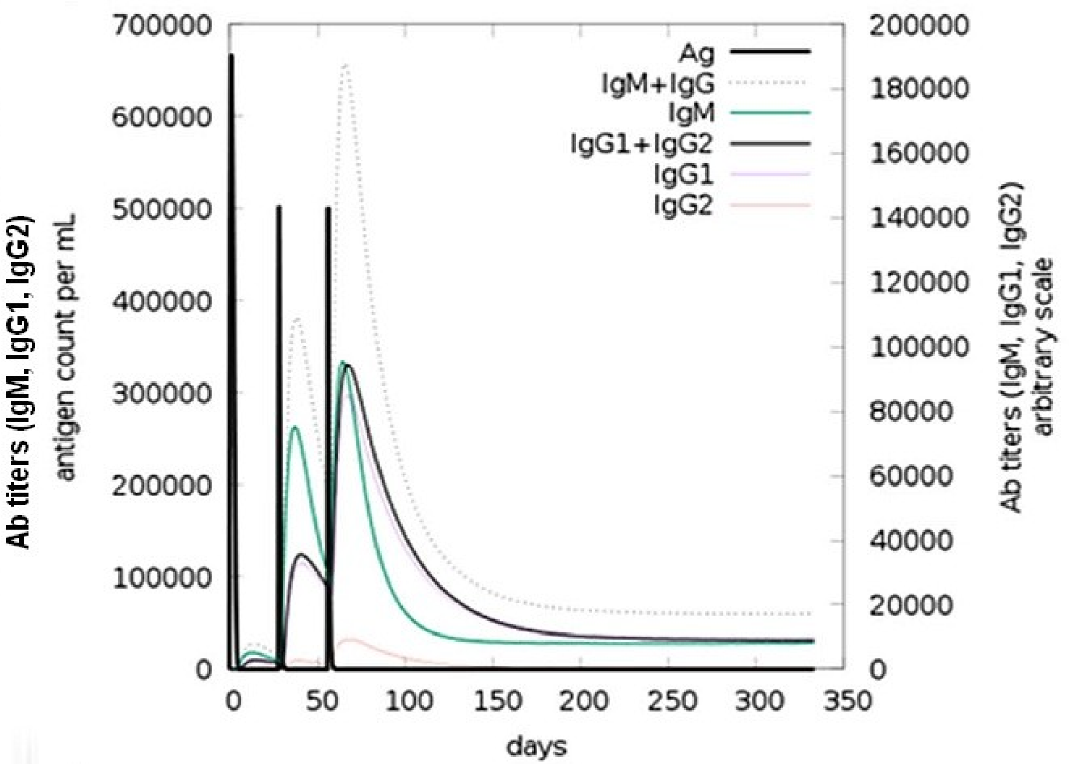




* Black = Antigen;

# Literature Review: CoV-2 / in-silico

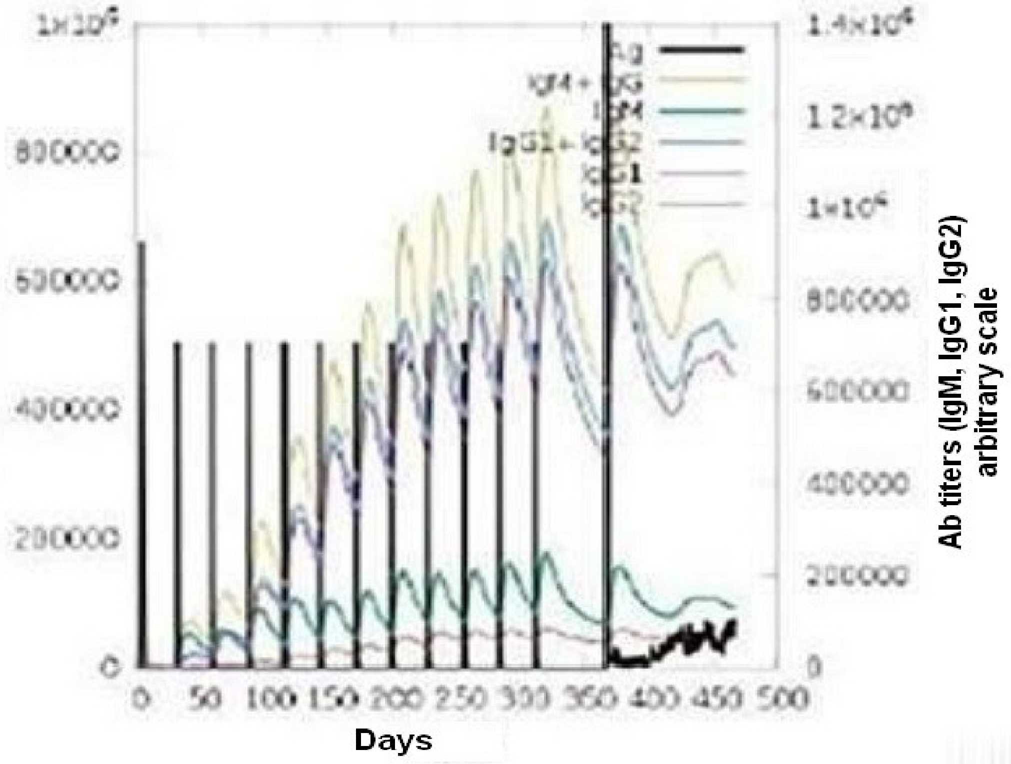
## Immune Simulations: Antibodies after 12 x Vaccin!



# Literature Review: CoV-2 / in-silico

## Immune Simulations: Antibodies after 12 x Vaccin!




**Only Pre-print:**

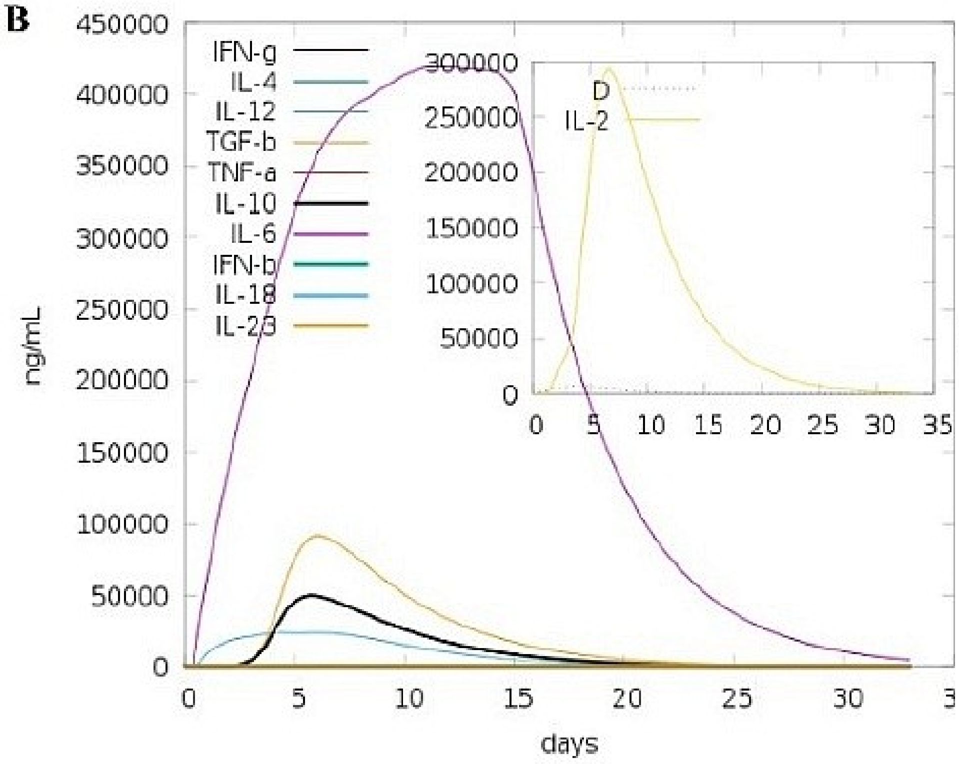
- 12 x Vaccine;
- Viral challenge afterwards;

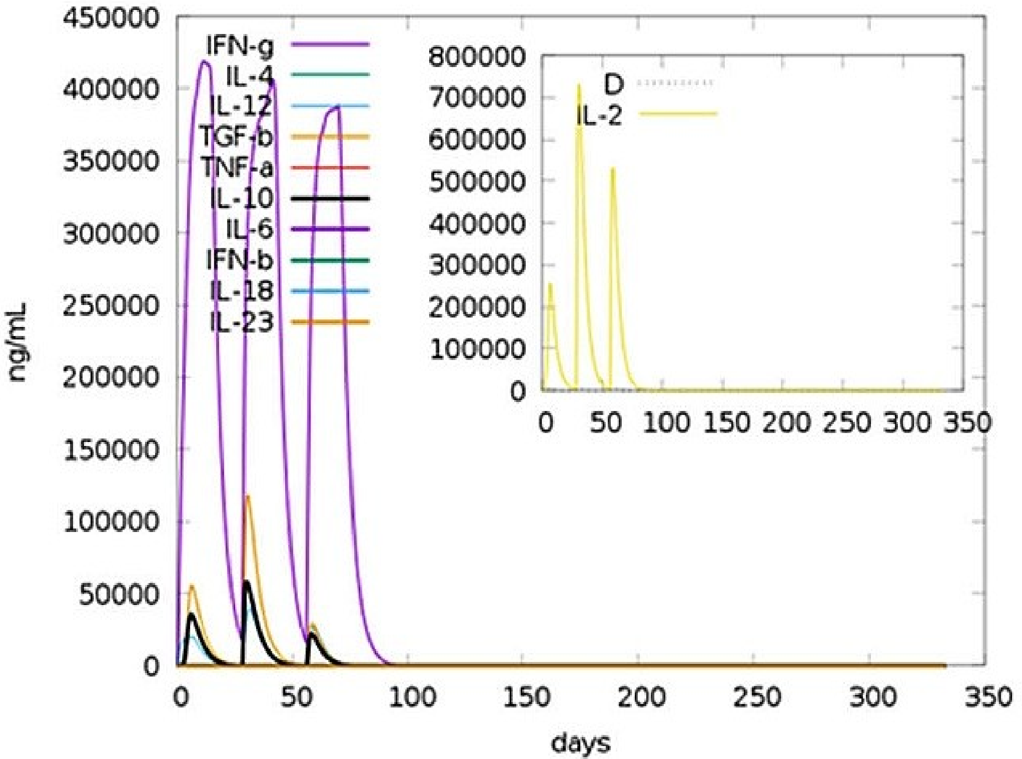
Immuno-informatics approach for multi-epitope vaccine designing against SARS-CoV-2. bioRxiv 18-Aug-2020.

https://doi.org/10.1101/2020.07.23.218529;

# Literature Review: CoV-2 / in-silico

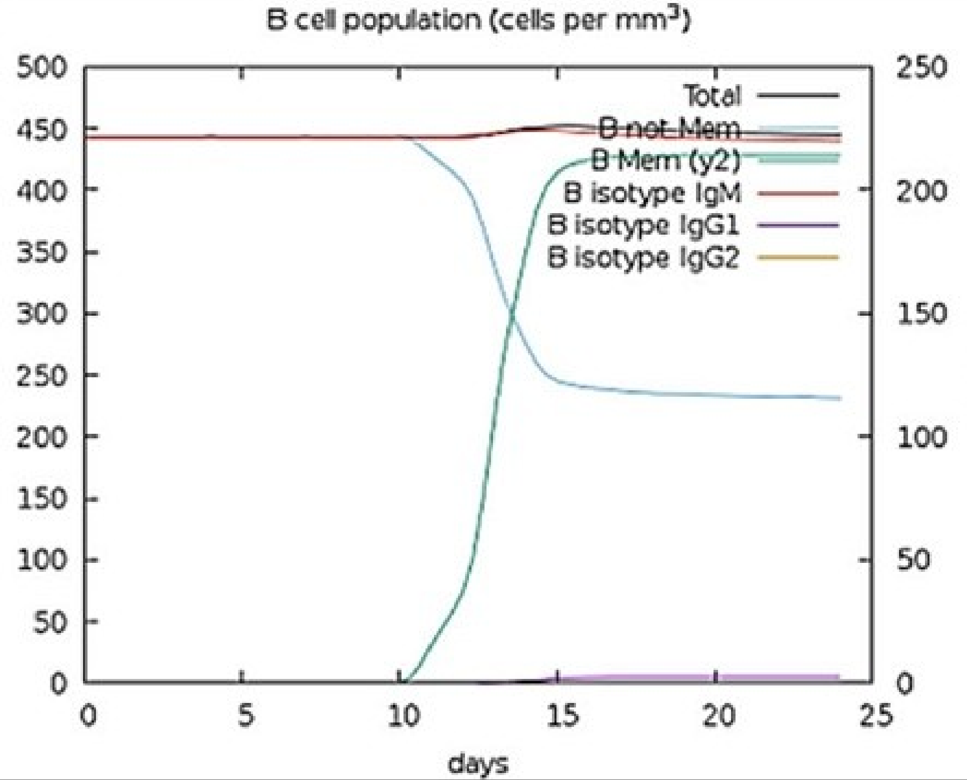
## **Immune Simulations: Cytokines**

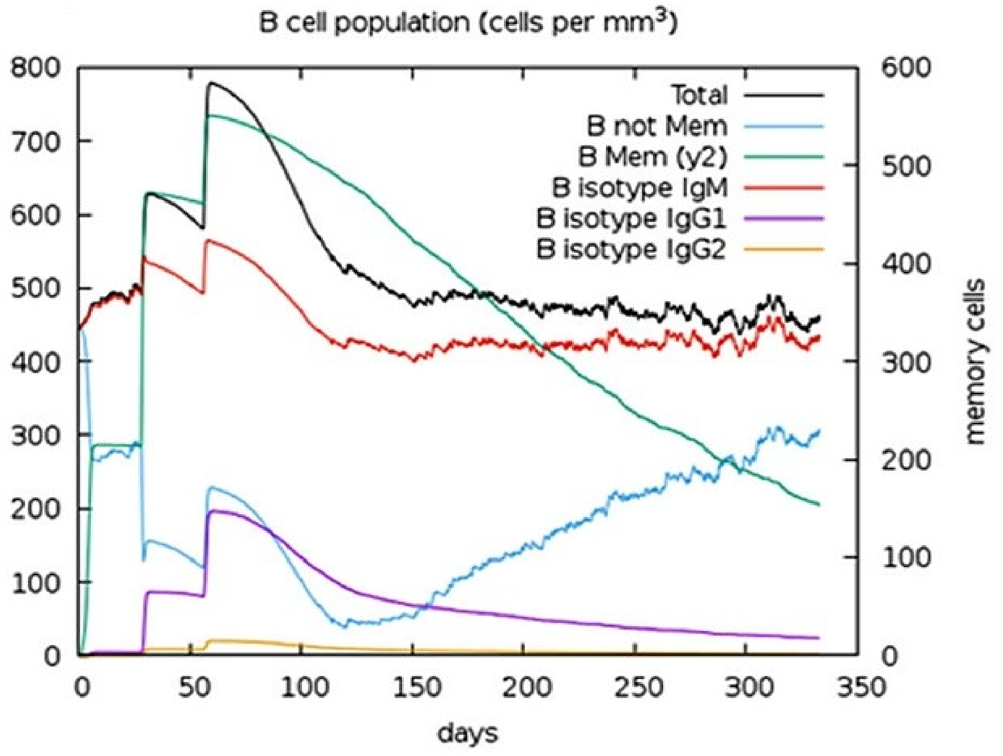


# Literature Review: CoV-2 / in-silico

## Immune Simulations: B Cells



# Pharmacology: CoV-2

- **Pharmacology:** Majority of PubMed articles with “**in-silico**”

# Pharmacology

**PubMed Articles:**

- 2020: 339 + most of remaining 70;
- Drug-Repurposing & Drug-Libraries;
- Phytocompounds;

**Molecular Docking:**

- Almost ALL articles;

# Pharmacology

**PubMed Articles:**

- 2020: 339 + most of remaining 70;

Summary: based on ~ 50% of articles:

- Phytocompounds: 240; (*)
- Plants: 61; (*)
- Drugs: 140;

* Final number may be x 2;

# Pharmacology: Plants

## Plants & Phytocompunds:

- 2020: 339 + most of remaining 70;
- Phytocompounds: 240;
- Plants: 61;
- Drugs: 140;

| | |
|---|---|
| Acacia senegal | Calendula officinals |
| Ammoides verticillata | Callophycus oppositifol. |
| Andrographis paniculata | Camellia sinensis |
| Annona muricata | Capsicum annuum |
| Artemisia annua | Castanosperm. australe |
| Asparagus racemosus | Caulerpa racemosa |
| Astragalus membranac. | Chenopodium quinoa |
| Brassica napus | Clerodendrum volubile |
| Broussonetia papyrifera | Crepis sancta |

* Summary: based on ~ 50% of articles; final number may be x 2;

# Pharmacology: Phytocompunds

## Plants & Phytocompunds:

- 2020: 339 + most of remaining 70;
- Phytocompounds: 240;
- Plants: 61;
- Drugs: 140;

| | |
|---|---|
| iridoid | kazinol |
| isatin | lauruside |
| isoginkgetin | leupeptin |
| isoliquiritigenin | limonene |
| isomesuol | limonoid |
| jacalin | lividomycin |
| jamaicin | lupane |
| kaempferol | macaflavanone |
| kamalachalcone | maslinic acid |

* Summary: based on ~ 50% of articles; final number may be x 2;

# Pharmacology: Targets

**Targets:**

- Main Protease (Mpro): > 124;
- CL-Protease: alone or in combination (few);
- Polymerase (RdRp): > 11;
- S or S/ACE2 complex: > 45;
- Other: Methyltransferase, etc.